GGG-S-283a
May 1, 1967

SUPERSEDING
Int. Fed. Spec. GGG-S-00283 (GSA-FSS)
April 8, 1964

FEDERAL SPECIFICATION

SHEARS, METAL CUTTING, PNEUMATIC (PORTABLE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers one type of portable pneumatic shears used for cutting sheet metal, plastics, etc.

1.1.1 Federal specification coverage. Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, or request for proposal, form a part of this specification.

Federal Specifications:

PPP-B-566—Boxes, Folding, Paperboard.
PPP-B-585—Boxes, Wood, Wireboard.
PPP-B-591—Boxes, Fiberboard, Wood-Cleated.
PPP-B-601—Boxes, Wood, Cleated-Plywood.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636—Box, Fiberboard.
PPP-B-640—Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-B-665—Boxes; Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676—Boxes, Setup.

Federal Standards:

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 151—Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-P-116—Preservation, Methods of.

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Military Standards:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.
MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following publications form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

National Bureau of Standards Handbook:

H28—Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402).

United States of America Standards Institute:

ASA B 3.5—Tolerances for Ball and Roller Bearings.
ASA B 3.6—Boundary Dimensions for Ball and Roller Bearings.
ASA B 3.8—Mounting Ball and Roller Bearings, Specifications For.

(Application for copies should be addressed to the United States of America Standards Institute, 10 East 40th Street, New York 16, N. Y.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.)

3. REQUIREMENTS

3.1 Material. Shears shall be of the best material that is entirely suitable for the intended use.

3.2 Design. The shears shall be of the manually held and guided type, with a pneumatic motor driving a suitable mechanism to produce a shearing action by a set of blades.

3.3 Construction.

3.3.1 Illustration. The illustration shown herein is descriptive and not restrictive, and is not intended to preclude the purchase of shears which are otherwise in accordance with the requirements of this specification.

3.3.2 General construction. The shears shall be simple in construction and reliable in operation at air pressure of 80 to 100 p.s.i., and shall have the parts readily accessible for inspection, adjustment, or replacement. The mechanisms shall be enclosed and thoroughly protected by metal cases. Working parts exposed to wear shall be of sufficient hardness to withstand the service required. Parts shall be properly finished by machining or grinding to dimensions within limits established by good commercial practice. Similar parts shall be interchangeable and of good fit when assembled with relative parts of the same size, type, and make of shears. All parts shall be suitably supported, mounted, and secured. Blade-clearance adjustment shall be simple and easily accomplished. Blades shall be easily removable and easily reground without the use of special fixtures. All threaded parts shall be in accordance with the applicable requirements of Handbook H28.

3.4 Strainer. Shears shall be fitted with an air strainer or screen of 20 mesh or finer which shall be effective in retaining solid particles in the compressed air supply. The strainer shall be of such design that it can be readily removed for cleaning.

3.5 Air inlet connection. The air inlet connection shall consist of well-secured ferrous fitting threaded with internal NPT- or NPTF threads conforming to Handbook H28.

3.6 Throttle. Shears shall be provided with a nonleakable, trigger-style, throttle for admitting and shutting of the supply of compressed air.

3.7 Rotor. The rotor shall be propelled by compressed air. The rotor proper shall consist of one or more elements properly balanced with the rotor shaft.

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3.8 Bearings. The rotor shaft and spindle shaft shall be mounted on ball or roller bearings. Other rotating parts shall be provided with commercially available bearings. Roller bearings which are designed primarily for radial loads shall not be used as thrust bearings. Ball and roller bearings shall be of the dustproof type or shall be mounted in a dustproof housing. Ball and roller bearings shall conform to Standard B 3.5 for tolerances, B 3.6 for dimensions, and B 3.8 for mounting requirements for the type and size bearings used. If sleeve bearings are used they shall be the sintered, metal powder, oil impregnated type.

3.9 Gears and shafts. Gears shall be of suitable heat-treated steel. All gears shall have cut teeth and be adequately supported. Shafts shall be of suitable alloy heat-treated steel. Each shaft shall be of the proper diameter to prevent deflection and shall be supported by at least two suitable bearings of the type specified in 3.8. Gears and shafts shall be fully enclosed.

3.10 Lubrication.

3.10.1 Primary lubrication. Shears shall be provided with an oil reservoir of sufficient capacity in one filling to supply lubrication to the rotor, cylinder, and other moving parts in the path of the air stream, for at least 8 hours of actual operation. The oil shall be fed from the reservoir during operation of the shears. The oil flow shall be regulated so that lubrication will be sufficient without excessive quantities of lubricant being discharged from the shears with the exhaust air.

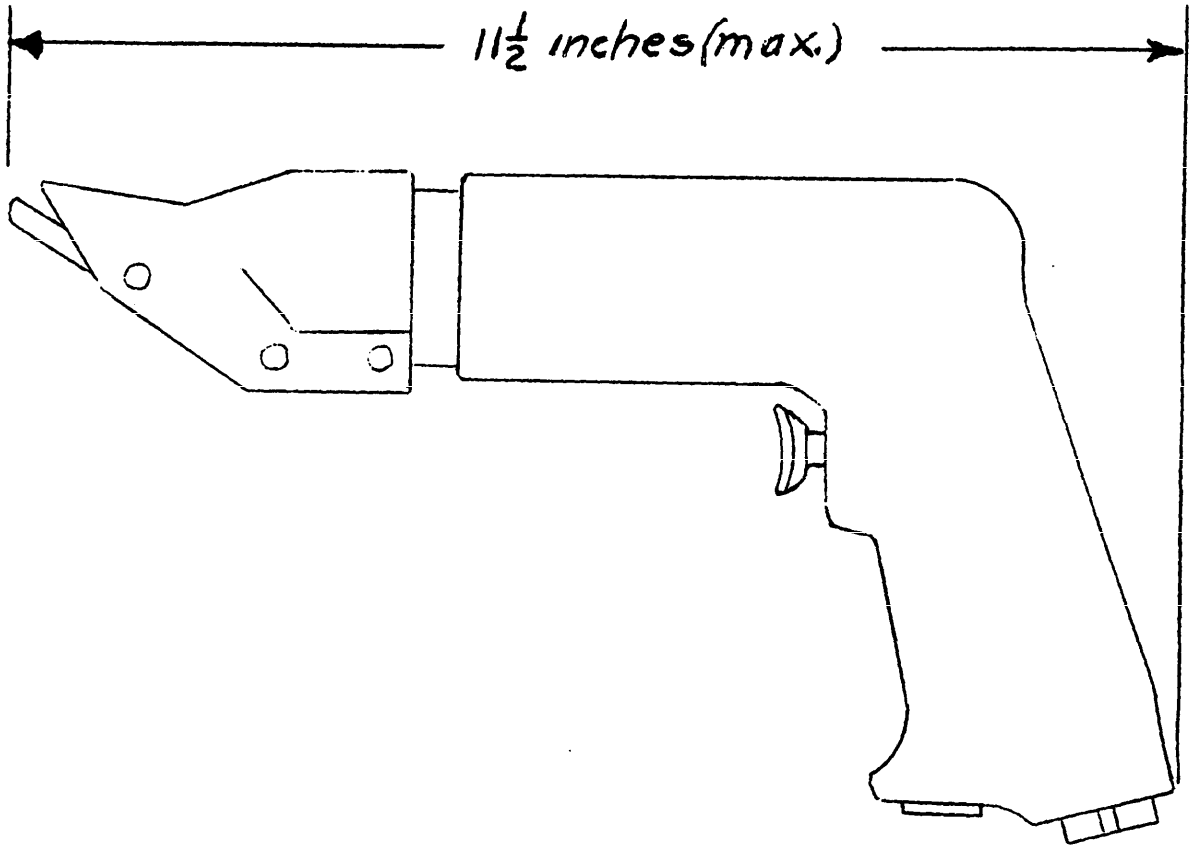


Figure 1. Shear, pneumatic, portable.

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3.10.2 Secondary lubrication. Conventional type lubricating fittings or openings or other suitable means shall be provided for introduction of sufficient lubricant for bearings, gears, and other moving parts not lubricated from the oil reservoir. Where acceptable prelubricated sealed bearings are used, no external means of lubrication shall be required for these parts.

3.11 Cutting characteristics. Shears shall be suitable for inside and outside cutting and shall be capable of cutting straight, irregular contours, and curves in sheet steel, and shall meet the cutting requirements of table I. They shall make a clean cut, without burrs or ragged edges, and shall not distort or mutilate the working part of the sheet. The cutting blades shall be clearly visible at all times to enable the operator to follow the line to be cut. The cutting blades shall be made of an alloy tool steel and heat treated to a hardness of 57-60 on the Rockwell C scale.

3.12 Detail requirements. The shears shall have one cutting blade, pivoted to oscillate between two stationary cutting blades to double shear a strip no wider than ¼ inch. The shears shall have a pistol grip handle, be similar to figure 1, and conform to table I.

TABLE I. Detail requirements

Size	Air inlet connection pipe size	Air consumption (max CFM		Cutting capacity, hot rolled SAE 1010 steel or equal		Cutting speed, minimum	Cutting radius minimum	Weight maximum
		at cutting test rate	at no load					
				Gage number	inch	Ft./Min	Inches	Pounds
1	¼	25	30	18	0.0490	15	3	4

3.13 Identification marking. Shears shall be provided with suitable identification showing the manufacturer's name, model or serial number and gage capacity for SAE 1010 hot rolled steel.

3.14 Handbook of instructions. The contractor shall furnish one copy of a handbook of instructions. Unless otherwise specified (see 6.1), the size and contents of the hand book shall be that normally furnished by the contractor in general commercial practice.

3.15 Interchangeable parts. All parts having the same manufacturer's part number shall be constructed to definite standards, tolerances, and clearance in order that such parts may be completely interchangeable, replaced or adjusted without requiring modification.

3.16 Workmanship. The workmanship shall be first-class commercial quality in every respect and free from all imperfections which may affect the serviceability of the shears.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components. In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used were manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2 Sampling procedures. Sampling procedures shall be in accordance with MIL–STD–105. Data for sampling shall be as stated in table II.

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TABLE II. Sampling Data

Category	Sample unit	Inspection level	Acceptable quality level	AQL expressed in terms of	Reference
Visual examination	1 ea.	II	4.0	Per cent defective	4.3.1
Dimensional and weight examination	1 ea.	II	4.0	Per cent defective	4.3.2
Testing	1 ea.	S–4	4.0	Per cent defective	4.4
Preparation for delivery	one container	S–2	4.0	Defects per hundred units	4.5

4.3 Examination.

4.3.1 Visual examination. Each sample unit shall be examined for any non-conformance in design, material, finish, construction, workmanship, and marking.

4.3.2 Dimensional and weight examination. Each sample unit shall be examined for any nonconformance with dimensional and weight requirements.

4.4 Testing. Each sample unit shall be tested in accordance with 4.4.1, 4.4.2, and 4.4.3.

4.4.1 Operational test. Shears shall be tested to determine compliance with 3.11 and table I.

4.4.2 Test conditions. All tests shall be based on operation of the shears with compressed air at a gage pressure of 90 p.s.i., measured at the tool. Test shears shall be connected to the air supply by means of an 8 foot length of hose provided at the tool end with a fitting mating with the air inlet connection specified in table I.

4.4.3 Hardness test. Shears shall be tested for the hardness requirements of 3.11 using method 243.1 of Fed. Test Method Std. No. 151.

4.5 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the preservation, packaging, packing and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table III. The sample unit shall be one shipping container fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot.

Table III.

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing, or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A, B or C as specified (see 6.1).

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5.1.1 Level A. The shears shall be preserved and packaged in accordance with method IA–8 of MIL–P–116 utilizing an outer container conforming to PPP–B–636, W6c. The box shall be closed in accordance with the appendix to the box specification.

5.1.2 Level B. The shears shall be packaged in a close fitting box conforming to PPP–B–566, PPP–B–665 or PPP–B–676. Cushioning shall be provided.

5.1.3 Level C. The shears shall be preserved and packaged to afford protection against corrosion, deterioration and damage from the supplier to the initial destination. The supplier may use his commercial practice providing it fulfills the above requirements.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. Unit packages of shears of like description in quantities as specified (see 6.1) shall be packed in a closs-fitting box conforming to PPP–B–585, class 3, PPP–B–591, class II, PPP–B–601, type III, PPP–B–621, class 2, PPP–B–636, class weather-resistant or PPP–B–640, class 2, grade A. The boxes shall be closed and strapped in accordance with the applicable specification or appendix thereto. The gross weight of the triple wall box or the wood boxes shall not exceed 200 pounds. The gross weight of the PPP–B–636 box shall not exceed the weight limitations of the box specification.

5.2.2 Level B. Unit packages of shears of like description in quantities as specified (see 6.1) shall be packed in a closs-fitting box conforming to PPP–B–636, class domestic. The box shall be closed in accordance with the appendix to the box specification. The gross weight shall not exceed the weight limitations of the box specification.

5.2.3 Level C. The shears shall be packed to insure carrier acceptance and safe delivery at destination in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to markings required by the contract or order, the packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military activities. In addition to markings required by the contract or order, the packages and shipping containers shall be marked in accordance with MIL–STD–129.

6. NOTES

6. Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Instruction book requirements, if different (see 3.14).
(c) Levels of preservation and packaging, and packing required (see 5.1 and 5.2).
(d) Marking required (see 5.3).

MILITARY CUSTODIANS:

Army—GL
Navy—SH
Air Force—SR

Review activities:

Navy—SH
Air Force—SR

User activities:

Army—GL
Navy—SH, YD

Military coordinating activity:

Navy—SH

Preparing activity:

GSA—FSS

U.S. Government Printing Office: 1967

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